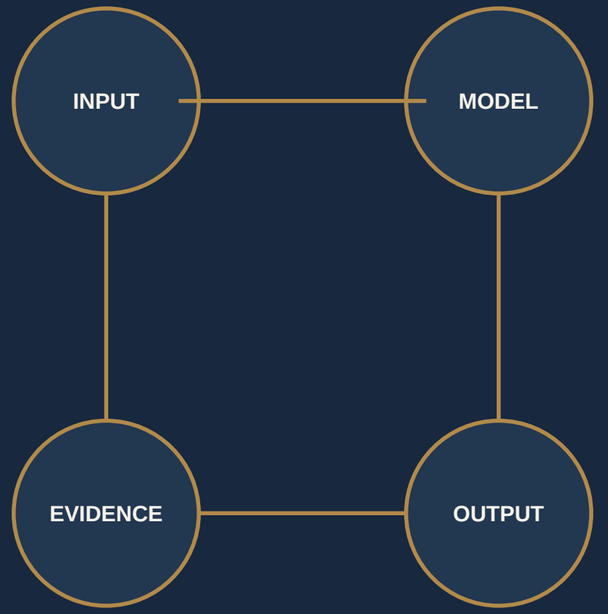
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Biologically Inspired Robotics

AI.306 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define locomotion and intelligence inspired by insects.
- Apply key methods including birds.
- Build or analyze a practical system involving fish.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Locomotion and intelligence inspired by insects

02

METHODS

Birds

03

SYSTEMS

Fish

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Locomotion and intelligence ir

Locomotion and intelligence inspired by insects

02 · Birds

Birds

03 · Fish

Fish

04 · Mammals and humans

Mammals and humans

05 · Mandatory practical: Design c

Mandatory practical: Design or simulate a biologically inspired robotic mechanism

06 · ### Year 3 — Semester 2

Year 3 — Semester 2

Core ideas, made explicit

AI.306 • Distinguish definitions, assumptions and evidence

- Locomotion and intelligence inspired by insects
- Birds
- Fish
- Mammals and humans
- Mandatory practical: Design or simulate a biologically inspired robotic mechanism

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.306 • Distinguish definitions, assumptions and evidence

- ### Year 3 — Semester 2

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving locomotion and intelligence inspired by insects.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using locomotion and intelligence inspired by insects.
- Implement or simulate a baseline using birds.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.